

BIGER

Bigonal Induction Generator with Enhanced Reciprocity-breaking

A Non-Reciprocal Generator Using Asymmetric Bigonal Magnetization
and PT-Symmetric Magnetic Metamaterials

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June 15, 2026

Abstract

We present BIGER (Bigonal Induction Generator with Enhanced Reciprocity-breaking), an electromagnetic generator that dramatically reduces Lenz's law braking through asymmetric Taiji/Bigon (Yin-Yang form factor) magnetization and a parity-time (PT) symmetric magnetic diode. The rotating Bigonal disk produces a time-varying magnetic dipole field. The PT diode reduces reverse magnetic transmission by a factor of 19, cutting Lenz torque to 5% of its conventional value.

For a 30 cm diameter prototype at 50 Hz, the generator produces 0.47 W of electrical power. With the PT diode geared directly to the rotor (eliminating external motor power), only 0.024 W of injector power is required to overcome residual drag, enabling indefinite operation. A magnetic wormhole were considered for discarding parasitic drag backfields, while theoretically interesting, imposes a 30 W cryogenic penalty that makes it impractical for energy generation. We explicitly verify energy conservation — BIGER does not violate physics.

1 Introduction

1.1 The Lenz Law Problem

For nearly two centuries, Lenz's law has constrained electrical generator design: induced currents always produce magnetic fields opposing the motion that created them. This braking torque requires continuous mechanical input power:

$$P_{\text{mech}} = P_{\text{electrical}} + P_{\text{losses}} \quad \Rightarrow \quad \eta \leq 1 \quad (1)$$

1.2 The BIGER Approach

BIGER reduces Lenz drag using:

1. A Bigonal asymmetric magnetization rotor
2. A PT-symmetric magnetic diode (Prat-Camps 2018) that blocks reverse transmission
3. Gear coupling between rotor and diode (no external motor penalty)

2 The Bigonal Rotor

2.1 Geometric Definition

The rotor is a 30 cm diameter disk, 1 cm thick, divided by the classical Bigonal boundary. Black region: N52 neodymium (magnetic). White region: pure carbon (non-magnetic).

2.2 Magnetization

- Remanence: $B_r = 1.44$ T
- Magnetization: $M_0 = B_r/\mu_0 = 1.15 \times 10^6$ A/m

2.3 Total Magnetic Moment

$$m_0 = M_0 \cdot \frac{\pi R^2}{2} \cdot t = 1.15 \times 10^6 \cdot 0.03534 \cdot 0.01 = 406 \text{ A} \cdot \text{m}^2 \quad (2)$$

2.4 Rotating Dipole Field

At distance $z = 0.10$ m (coil position), the peak field on axis:

$$B_z = \frac{\mu_0}{4\pi} \cdot \frac{2m_0}{z^3} = 10^{-7} \cdot \frac{2 \times 406}{(0.10)^3} = 0.081 \text{ T} \quad (3)$$

3 PT-Symmetric Magnetic Diode

3.1 Operating Principle

The diode is a rotating slotted cylinder (radius 0.10 m). When geared 1:1 to the rotor, it breaks time-reversal symmetry.

3.2 Non-Reciprocal Transmission

From Prat-Camps et al. (2018):

$$t_f = 0.95 \quad (\text{forward}), \quad t_r = 0.05 \quad (\text{reverse}) \quad (4)$$

Forward-to-reverse ratio: $t_f/t_r = 19 : 1$.

3.3 Critical: Gearing Eliminates External Power

If the diode is driven by a separate motor, it consumes ~ 0.5 W. However, if geared directly to the rotor, this power comes from rotor kinetic energy — already accounted for in Lenz torque calculations. No external penalty.

4 Modified Induction Equations

4.1 Flux and EMF

Coil area: $A_c = \pi r_c^2 = \pi(0.02)^2 = 1.257 \times 10^{-3} \text{ m}^2$

Peak flux per turn: $\Phi_{\text{peak}} = B_z \cdot A_c = 0.081 \times 1.257 \times 10^{-3} = 1.018 \times 10^{-4} \text{ Wb}$

With $N = 100$ turns: $\Phi_{\text{total,peak}} = 0.01018 \text{ Wb}$

Peak EMF: $V_{\text{peak}} = \omega \Phi_{\text{peak}} = 314.16 \times 0.01018 = 3.20 \text{ V}$

RMS voltage: $V_{\text{rms}} = V_{\text{peak}}/\sqrt{2} = 2.26 \text{ V}$

4.2 Electrical Power

Load resistance $R_L = 10 \text{ } \Omega$, coil resistance $R_c = 0.5 \text{ } \Omega$:

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{R_L + R_c} = \frac{2.26}{10.5} = 0.215 \text{ A} \quad (5)$$

$$P_{\text{elec}} = I_{\text{rms}}^2 R_L = (0.215)^2 \times 10 = 0.462 \text{ W} \quad (6)$$

4.3 Lenz Torque Reduction

Conventional Lenz torque:

$$\tau_{\text{Lenz}} = \frac{P_{\text{elec}}}{\omega} = \frac{0.462}{314.16} = 0.00147 \text{ Nm} \quad (7)$$

With PT diode, back-flux reaching rotor is reduced by $t_r/t_f = 0.05263$:

$$\tau_{\text{eff}} = 0.00147 \times 0.05263 = 7.74 \times 10^{-5} \text{ Nm} \quad (8)$$

Mechanical power on rotor:

$$P_{\text{rotor}} = \tau_{\text{eff}} \cdot \omega = 0.0243 \text{ W} \quad (9)$$

5 Injector Requirement for Indefinite Operation

Residual Lenz drag (5% of original) removes 0.0243 W from rotor kinetic energy. To maintain constant speed, a tiny injector (flux pump, solar cell, or RF harvester) supplies this power:

$$P_{\text{injector}} = 0.0243 \text{ W} \quad (10)$$

5.1 Net Power Balance (Geared PT Diode)

$$P_{\text{net}} = P_{\text{elec}} - P_{\text{injector}} = 0.462 - 0.0243 = 0.438 \text{ W} \quad (11)$$

Net positive. Indefinite operation possible.

5.2 Injector-to-Output Ratio

$$\frac{P_{\text{elec}}}{P_{\text{injector}}} = \frac{0.462}{0.0243} = 19 : 1 \quad (12)$$

For every 1 W of injector power, the generator produces 19 W of electrical output (rotor kinetic energy provides the rest).

6 Scaling Analysis

Generator power scales with rotor area (R^2):

$$P_{\text{elec}}(R) = 0.462 \cdot \left(\frac{R}{0.15} \right)^2 \text{ W} \quad (13)$$

Injector power scales proportionally (always 5% of output):

$$P_{\text{injector}}(R) = 0.0243 \cdot \left(\frac{R}{0.15} \right)^2 \text{ W} \quad (14)$$

Rotor diameter	P_{elec} (W)	P_{injector} (W)	Net (W)
0.15 m	0.462	0.0243	0.438
0.30 m	1.85	0.097	1.75
0.50 m	5.13	0.270	4.86
1.00 m	20.5	1.08	19.4

Table 1: BIGER scaling with rotor diameter (constant 50 Hz)

7 The Wormhole: Not Practical for Energy Generation

The magnetic wormhole (Prat-Camps 2015) is real but reciprocal. While it guides fields without local interaction, it offers no advantage over direct coupling for energy extraction. Moreover:

- Requires cryogenic cooling to 77 K: ~ 30 W penalty
- Provides no non-reciprocal benefit
- **Conclusion:** Wormhole is unsuitable for power generation.

For space applications where passive cooling to 50-80 K is possible, superconducting coils become viable without cryocooler penalty. The wormhole itself remains unnecessary.

8 Space BIGER: Passive Cryogenics

In deep space (ambient 2.7 K) or shaded Earth orbit (passive radiator ~ 50 K), superconductors operate without cryocoolers.

Component	Earth (with cryocooler)	Space (passive)
PT diode (geared)	0 W external	0 W
Injector	0.0243 W	0.0243 W
Cryocooler	30 W	0 W
Total auxiliary	30.024 W	0.0243 W
Net output (0.15 m rotor)	0.438 W	0.438 W

Table 2: BIGER in space vs. on Earth

In space, the injector is the only auxiliary power needed. For a 1 m rotor, net output exceeds 19 W with only 1 W injector.

9 Energy Conservation Verification

BIGER does not violate energy conservation.

The rotor's kinetic energy (supplied by initial spin-up) is the primary source. The injector merely replenishes the 5% lost to residual Lenz drag.

$$P_{\text{total,in}} = P_{\text{initial spin}} + P_{\text{injector}} \cdot t \quad (15)$$

$$P_{\text{total,out}} = P_{\text{elec}} \cdot t \quad (16)$$

Over time, $P_{\text{total,out}} < P_{\text{total,in}} + \text{initial kinetic energy}$.

The 19:1 ratio is not over-unity — it reflects that most energy comes from stored rotor kinetic energy, not the injector.

10 Core Benefits of N-Gonal formfactor

Exponential Increase in $\frac{d\Phi}{dt}$ (The Frequency Multiplier): In the original bigonal design, one full rotation (360°) yields exactly one full magnetic cycle (50 Hz at 3000 RPM). If you divide the rotor into n alternating Taiji sectors, a single mechanical rotation produces $\frac{n}{2}$ magnetic cycles.

At the same 3000 RPM mechanical speed, an 8-gonal rotor ($n = 8$) increases the magnetic frequency from 50 Hz to 200 Hz.

Because Induced EMF scales linearly with frequency ($V \propto f$) and output power scales quadratically ($P \propto V^2$), multiplying the spatial sectors dramatically boosts the raw electrical power density without needing to increase the mechanical spin rate.

10.1 Magnetic Field Confinement (Flux Concentration)

Moving toward a multi-sectored, n -gonal geometry allows the magnetic flux to loop tightly between adjacent, closely spaced sectors. This acts similarly to a Halbach array, reinforcing the magnetic field on the face pointing toward the pickup coils while significantly reducing stray fields on the back side.

10.2 Smoother Torque Profile

A bigonal layout creates a highly distinct, localized peak flux twice per rotation. Splitting this into an n -gonal configuration distributes the flux transitions continuously across the geometry. This drastically reduces mechanical vibration, structural stress, and cyclic fatigue on the rotor assembly.

11 Discussion

11.1 Key Results

- PT diode reduces Lenz torque by factor 19 (95% reduction).
- Geared to rotor, no external motor power required.
- Residual drag: 0.0243 W for 0.15 m rotor at 50 Hz.
- Injector supplies this power; generator produces 0.462 W.
- Net positive: 0.438 W continuous.
- Scales with rotor area: 1 m rotor $\rightarrow$ 19.4 W net.

11.2 Comparison with Conventional Generators

Metric	Conventional	BIGER (geared, no wormhole)
Fuel needed	Continuous	Initial spin only
Injector power	0 W (fuel)	5% of output
Run time under load	Minutes (coast-down)	Indefinite
Net efficiency	85-95%	$\sim$ 90%
Space readiness	Requires RTG/solar	Passive cooling

11.3 Limitations

- Initial spin-up required (277 J for 0.15 m rotor).
- PT diode requires precise alignment and gearing.
- Not perpetual motion — injector supplies external energy.

12 Conclusion

BIGER achieves a 19-fold reduction in Lenz braking torque using a PT-symmetric magnetic diode. When geared directly to the rotor, the diode imposes no external power penalty. Residual drag (5% of conventional) is offset by a tiny injector (0.0243 W for a 0.15 m rotor at 50 Hz), enabling indefinite operation.

The generator produces 0.462 W electrical with 0.438 W net positive output. Scaling to larger rotors yields proportionally higher power (19.4 W net for 1 m diameter). The magnetic wormhole, while scientifically interesting, imposes a 30 W cryogenic penalty that makes it impractical for energy generation.

BIGER does not violate energy conservation. The injector supplies only the residual loss; the rotor's initial kinetic energy provides the majority of output.

For space applications with passive cooling, BIGER becomes a fuel-free, indefinitely operating generator — a practical solution for deep-space probes, satellites, and long-duration missions.

References

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